

Do Tran Hieu

INTERN JAVA BACKEND

dotranhieu925@gmail.com | 0969 792 520

Career Objective

Java Backend Intern specializing in Spring Boot and RESTful APIs, with experience building real-time and scalable applications. Strong in concurrency handling, system design, and integrating tools like Elasticsearch and Docker. Eager to contribute to production systems and grow in backend engineering.

Education

Posts and Telecommunications Institute of Technology (PTIT)

2023 – Present

Major: Computer Science

GPA: 3.44 / 4.0 CPA: 3.47 / 4.0

Honors & Awards

Academic Excellence Scholarship (Excellent Category)

Semester 4, 2025 – 2026

Awarded by PTIT for outstanding academic performance (GPA-based merit scholarship).

Entrance Scholarship

The first year 2023 - 2024

Awarded Entrance Scholarship based on academic excellence.

Skills

Programming Languages: Java, C++, Python, JavaScript (basic)

Backend Development: Spring Boot, RESTful APIs, MyBatis(SQL Mapper), Spring Security (JWT), JPA/Hibernate, Web-Socket, Kafka(basic), Spring Cloud, Spring AOP, Spring Scheduled

Search Engine: Elasticsearch

Security: OAuth2 Login(Google, Facebook), Firebase(Authen)

Frontend: ReactJS, Angular(basic)

Database & Migration: MySQL, SQL Server, PostgreSQL, Redis, Flyway

Architecture: Monolithic, Microservices

Cloud & Storage: Cloudinary, MinIO

DevOps & Tools: Git/GitLab, Swagger UI, Maven, Postman, Docker, Azure, CI/CD(Github Action), Consul

Work Experience

Backend Developer Intern

Mar 2026 – May 2026

DEHA Vietnam

Task Management Website

GitLab: gitlab.com/dotranhieu925/first-demo

Materialized rigorous, end-to-end training spanning Linux system administration, Git collaborative workflows, development, and Microsoft Azure deployment by independently engineering a real-time, scalable Task Management System as an internal internship project focusing on data consistency and performance.

Key Responsibilities & Achievements:

- Designed and implemented a relational data model supporting many-to-many relationships between users, teams, and tasks.
- Built real-time communication features (group chat, private messaging) using WebSocket.
- Integrated **Yjs** for collaborative document editing with active user tracking.
- Developed real-time commenting and notification system for task updates and assignments.
- Ensured data consistency in concurrent environments using **Optimistic Locking**.
- Improved search performance by integrating Elasticsearch for full-text search.
- Designed and exposed RESTful APIs secured with JWT authentication.

Technologies Used:

- Spring Boot, Spring Security, JPA/Hibernate, WebSocket
- ReactJS, Yjs
- MySQL, Elasticsearch (Dockerized)
- Docker, Maven, Git, Postman

Projects

HealthBooking

May 2026 - Present

GitLab: gitlab.com/dotranhieu925-group/hospital_booking

Developed a high-concurrency healthcare appointment booking platform connecting patients with hospitals and doctors, supporting three main user roles (Admin, Doctor, Patient). The system streamlines scheduling, automates slot generation, handles secure deposit payments, and features an AI-driven medical consultant.

Key Features & Technical Highlights:

- **Patient Side:** User registration and authentication; profile management; browsing departments, medical services, and doctor information; searching available appointment slots; booking appointments; online deposit payment via VNPAY; appointment tracking and history management.
- **Doctor Side:** Managing personal work schedules; registering and updating weekly availability; viewing assigned appointments and patient booking information; monitoring upcoming consultation sessions.
- **Admin Side:** Hospital administration including management of departments, medical services, doctors, and doctor schedules; monitoring appointment statistics and booking activities through administrative dashboards.
- **AI-Powered Consultation Chatbot:** Integrated Gemini API in **recommend-service** using dynamic database context (symptoms-to-departments mapping). Designed custom markdown redirect schemes parsed by React to guide patients directly from symptom triage to pre-filled booking checkouts.
- **Asynchronous System Auditing:** Built a custom, non-intrusive auditing system using **Spring AOP (Aspect-Oriented Programming)** that serializes request payloads, client IPs, user contexts, and dispatches log events asynchronously via **Apache Kafka** to a dedicated audit service.
- **Automated Scheduling:** Automated doctor availability slot generation using Spring Scheduled Tasks to run every Friday at 08:00 AM, creating slots only for the subsequent week to avoid schedule drift.
- **Payment & Refund Integration:** Integrated VNPAY payment gateway, supporting batch checkout transactions using group IDs and automated VNPay API refund processing for canceled appointments.

Technologies Used:

- **Backend:** Spring Boot 3.x, Spring Security, Spring Cloud Gateway, Spring Data JPA, OpenFeign, Spring AOP, Spring Scheduled
- **Middleware & Discovery:** Apache Kafka (KRaft), HashiCorp Consul (Service Discovery), Redis Cache, MinIO Object Storage
- **Architecture:** Microservices Architecture (Authn, Author, Profile, Doctor, Schedule, Booking, Payment, Recommend, Notification, Audit, and Base shared library)
- **Database:** PostgreSQL (Multi-schema per microservice design)

Interests

Artificial Intelligence, Machine Learning basics, Algorithms, Reading technical blogs